

ALANKARA ABHISHEK

Electronics / Application Engineer — IoT — Embedded Systems

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Professional Summary

Electronics engineering student with hands-on experience in IoT and embedded system development using ESP32, sensors, motor drivers, Bluetooth, and Arduino IDE. Experienced in developing hardware-software integrated systems, real-time monitoring solutions, and automation prototypes. Strong foundation in C and Python, with practical experience in sensor integration, motor control, Firebase connectivity, and real-time data acquisition. Interested in electronics engineering roles involving prototyping, robotics, embedded systems, circuit-oriented development, and testing.

Technical Skills

Embedded Systems: ESP32, Arduino IDE, Embedded C, Sensors, Bluetooth

IoT & Connectivity: Firebase, IoT Systems, Real-Time Monitoring

Programming: C, Python, SQL

Engineering Tools: MATLAB, Verilog

Hardware: Servo Motors, L298N Motor Driver, MAX30102, Temperature Sensor

Core Areas: Embedded Systems, Automation, Hardware-Software Integration, Prototyping, Testing

Research Experience

Research Intern – Machine Learning

Sep 2025 – Dec 2025

- Developed a hybrid GAT-LSTM deep learning model for industrial battery health prediction using real-world battery datasets.
- Achieved 98% prediction accuracy through model training and evaluation.
- Worked with battery-related datasets for health prediction and performance analysis.

Research Intern – Machine Learning

Jun 2025 – Aug 2025

- Built and evaluated a CNN-LSTM model for DNA sequence-based disease classification.
- Achieved 94% classification accuracy through model development and evaluation.

Projects

Smart Object Dispenser using IoT

- Developed an IoT-based automated dispensing system using ESP32 for real-time control and automation.
- Integrated servo motors and L298N motor driver for controlled mechanical operation.
- Implemented Firebase connectivity for remote monitoring and control.
- Programmed and tested the embedded system using Arduino IDE.

Wearable IoT-Based Health Monitoring Glove for High Altitude Safety

- Designed and developed a wearable IoT-based system for real-time monitoring of heart rate, SpO2, and body temperature.
- Integrated ESP32 with MAX30102, pulse sensor, and temperature sensor.
- Implemented Bluetooth connectivity and emergency alert functionality.
- Performed sensor integration and real-time data acquisition.

Education

Bachelor's Degree – SRM University AP

2023 – Present

CGPA: 7.71/10

Class XII – Sri Chaitanya Educational Institutions

2021 – 2023

Percentage: 95.7%

Class X – Andhra Pradesh Model School

2021

Percentage: 90%

Certifications

- **Free Embedded Systems Course** – July 2026
- **Cisco Networking Academy** – Operating Systems Basics, July 2026
- **IBM SkillsBuild** – Artificial Intelligence Fundamentals, July 2026